# Business Requirement Document

E-Commerce Order Management Database System

**Project Phase:** Week 1 – Understanding the Business Problem

**Role:** Database Analyst

**Document Type:** Business Requirement Document (BRD)

## 1. Purpose of This Document

This Business Requirement Document (BRD) translates the business scenario of the E-Commerce Order Management System into a formal set of functional requirements, non-functional requirements, project scope, and assumptions/constraints. It is intended to be the reference point against which the database design and later implementation is validated.

## 2. Functional Requirements

### 2.1 Customer Management

- The system shall allow a new customer to register with Name, Email, Mobile Number, and Address.
- The system shall record the Registration Date automatically at the time of sign-up.
- The system shall allow a customer's contact details and address to be updated.
- The system shall prevent duplicate customer records using a unique Customer ID and Email.

### 2.2 Category and Product Management

- The system shall allow products to be grouped under a Category with a Category Name and Description.
- The system shall allow creation, update, and retrieval of Product records including Product Name, Price, and Stock Quantity.
- The system shall associate every Product with exactly one Category.
- The system shall track and update Stock Quantity whenever a product is ordered.

### 2.3 Supplier Management

- The system shall maintain a record of Suppliers with Supplier Name and Contact Information.
- The system shall support associating products with the supplier(s) that provide them for restocking purposes.

### 2.4 Order Management

- The system shall allow a Customer to place an Order that records Order Date, Total Amount, and Order Status.

- The system shall allow an Order to contain one or more products through Order Details, each with Quantity and Unit Price.
- The system shall calculate the Total Amount of an order from its Order Details.
- The system shall allow the Order Status to be updated (for example: Placed, Confirmed, Shipped, Delivered, Cancelled).

### 2.5 Payment Management

- The system shall record a Payment against an Order with Payment Method, Payment Date, and Payment Status.
- The system shall support common payment statuses such as Pending, Completed, and Failed.
- The system shall ensure a Payment record cannot exist without a corresponding valid Order.

### 2.6 Shipment Management

- The system shall record Shipment details for an Order including Delivery Address, Shipment Date, and Delivery Status.
- The system shall allow Delivery Status to be tracked through stages such as Processing, Shipped, In Transit, and Delivered.

### 2.7 Review Management

- The system shall allow a Customer to submit a Review for a Product with a Rating and Comments.
- The system shall associate every Review with both the Customer who wrote it and the Product it refers to.

### 2.8 Reporting

- The system shall support generation of reports such as sales by category, best-selling products, order status summaries, and customer order history.
- The system shall allow filtering of reports by date range, product, or category.

## 3. Non-Functional Requirements

| Category | Requirement |
|---|---|
| Performance | The system should retrieve order, product, and customer information within an acceptable response time even as data volume grows. |
| Scalability | The database structure should accommodate a growing number of customers, products, and orders without redesign. |
| Data Integrity | The system must enforce referential integrity between related entities (e.g., an Order Detail cannot reference a non-existent Product or Order). |
| Security | Customer personal data and payment information must be protected from unauthorized access. |
| Availability | The system should be available for order placement and browsing during normal business operating hours with minimal downtime. |
| Reliability | Order, payment, and shipment data must remain consistent even in the event of partial transaction failures. |
| Usability | Data entry and retrieval structures should be designed so that reports and queries are straightforward to build. |
| Maintainability | The database schema should be well-documented and normalized to simplify future maintenance and enhancement. |
| Backup and Recovery | The system should support periodic backup of data to allow recovery in case of failure. |
| Auditability | Key transactions such as orders, payments, and status changes should be traceable through recorded dates and statuses. |

## 4. Project Scope

### 4.1 In Scope

- Customer registration and profile record-keeping.
- Product catalog management organized under categories.
- Supplier record-keeping for procurement reference.
- Order placement covering multiple products per order.
- Payment record-keeping linked to orders.
- Shipment and delivery status tracking linked to orders.
- Customer review and rating submission for products.
- Basic business reporting drawn from the above data (sales, order status, top products).

### 4.2 Out of Scope

- Real-time payment gateway integration (only payment status record-keeping is covered).
- Third-party courier/logistics API integration (only delivery status record-keeping is covered).

- Marketing, promotions, discount coupon, and loyalty program modules.
- Multi-language or multi-currency support.
- Mobile or web front-end application development (this project is limited to the database layer).

## 5. Assumptions and Constraints

### 5.1 Assumptions

- Each customer registers with a unique email address before placing an order.
- Each product belongs to exactly one category at a time.
- An order must contain at least one product (via Order Details).
- Only one payment record is expected per order under normal conditions.
- Only one shipment record is expected per order under normal conditions.
- A customer can only review a product after it has been part of an order they placed.

### 5.2 Constraints

- Only the ten predefined core entities and their listed attributes may be used; no entities or attributes may be added, removed, or renamed.
- The database design must remain consistent with these entities throughout the semester (ER modeling, normalization, SQL implementation, and reporting).
- The project timeline follows a weekly deliverable structure, requiring requirements to be finalized before design work begins.
- The solution must be implementable using standard relational database technology.